

HTTP - a Hybrid Transformer for Trajectory Predictor

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I. INTRODUCTION

Autonomous vehicles are widely considered to be a major part of the future of transportation. However, their deployment still faces important challenges, particularly in the detection and motion prediction of surrounding agents such as pedestrians and vehicles within complex driving environments.

Developing such systems requires both large-scale multimodal datasets collected from multiple sensors and robust machine learning models capable of understanding dynamic traffic scenes. In the context of this project, our group focused specifically on vehicle trajectory prediction in urban environments.

The obtained results are consistent with the existing literature on the subject. The following sections dive deeper into the different aspects of the project, including the dataset, model architecture, training procedure, and evaluation metrics.

II. STATE OF THE ART

This section presents the main concepts and architectures related to multimodal perception and trajectory forecasting for autonomous driving systems.

A. NuScenes Dataset

For this project, we relied on the nuScenes dataset [1], which consists of driving scenes recorded using a Renault Zoe electric vehicle equipped with several type of sensors, including cameras, lidar sensors, and radar sensors.

The dataset contains recordings collected in Boston and Singapore using a similar sensor configuration. In this work, we use the mini version of the original dataset containing 1000 scenes of 20 seconds recorded at a frequency of 2 Hz.

1) *Objective of the nuScenes dataset:* Reliable detection and tracking of surrounding agents are essential components of autonomous driving systems. To safely navigate complex urban environments, autonomous vehicles must continuously estimate the position and behavior of nearby road users such as pedestrians, cyclists, and vehicles.

Machine learning methods now play a central role in these perception tasks. However, although standard image datasets are highly accessible, they fail to fully capture the complexity of autonomous driving environments. As a result, multimodal datasets are generally preferred despite being significantly

more difficult and expensive to collect.

2) *Multimodal datasets:* No individual sensor can independently provide all the information required for reliable autonomous driving. For this reason, modern autonomous vehicles combine several complementary sensing technologies.

Cameras

- Advantage: cameras provide detailed visual information such as colors, textures, edges, and lighting conditions, enabling efficient object classification and accurate 2D perception.
- Disadvantage: cameras provide limited direct depth estimation and reduced accuracy for precise 3D localization.

Lidar point clouds

- Advantage: lidar sensors provide highly accurate 3D localization of surrounding objects.
- Disadvantage: lidar measurements are sparse and contain less semantic information than image data. Their detection range is also generally limited between 50 to 150 meters.

Radar sensors

- Advantage: radar sensors provide an extended detection range, typically between 200 and 300 meters, it also enables velocity estimation using Doppler effect.
- Disadvantage: radar measurements are even sparser and less spatially precise than lidar observations.

Because these type of sensors provide complementary information, combining them generally improves the robustness and reliability of perception systems in complex traffic environments.

3) *Annotations:* Objects present in the keyframes are manually annotated and divided into 23 semantic categories, including several vehicle classes, pedestrians, cyclists, and additional road users.

B. Decoupling Motion Forecasting

DeMo [6] proposes a trajectory forecasting framework for autonomous driving that aims to improve both the diversity and temporal consistency of predicted trajectories. Most recent forecasting methods rely on vectorized scene representations and follow a one-query-one-trajectory strategy, where each query directly predicts a single future trajectory. Although effective, this approach often provides limited modeling of the temporal evolution of vehicle states.

To overcome this limitation, DeMo introduces a decoupled query representation combined with State Space Models based on the Mamba architecture. Compared to conventional Transformer-based sequence modeling approaches, Mamba offers improved efficiency for long sequential dependencies while preserving strong temporal modeling capabilities.

The proposed framework separates trajectory representation into two complementary query types:

- **Motion mode queries**, which encode possible driving intentions such as lane following, left turns, or right turns.
- **Dynamic state queries**, which model the evolution of vehicle states over time and enforce temporal consistency across predicted trajectories.

The scene information is first extracted from HD maps and neighboring agents. Map polylines are encoded using a PointNet-based encoder, while historical trajectories are processed using several unidirectional Mamba blocks to capture temporal dependencies. These representations are then combined to build the final scene context features.

To jointly model motion intentions and temporal dynamics, the framework introduces three complementary modules:

- **State Consistency Module**: this module processes the dynamic state queries. The queries are first initialized through a Multi-Layer Perceptron (MLP), then refined through cross-attention interactions with the scene context features. Bidirectional Mamba layers are subsequently used to model temporal relationships between successive dynamic states.
- **Mode Localization Module**: this component focuses on directional intentions. Motion mode queries interact with the scene context through multi-head attention layers before MLP blocks estimate future trajectories and their associated confidence scores.
- **Hybrid Coupling Module**: this module combines motion mode queries and dynamic state queries in order to jointly model spatial intentions and temporal dynamics. Attention mechanisms are used to exchange information between the different query types before generating the final trajectory predictions.

The model is trained using a global loss function combining trajectory regression, classification losses, and auxiliary losses related to motion modes and temporal state consistency.

By separating directional intentions from temporal dynamic modeling, DeMo achieves a more detailed representation of future trajectories.

Experimental results obtained on the nuScenes and Argoverse 2 datasets show that this decoupled strategy improves trajectory prediction accuracy while maintaining efficient inference performance. This approach will serve as one of the main inspirations for our proposed model.

III. OUR MODEL

The HTTP model presented in this paper uses both LiDAR points and pictures taken from the front camera in order to predict the trajectory of nearby vehicles. The inputs are refined

using either the PointPillar model [4] for the LiDAR, or the ResNet-18 model [3] to process the pictures.

The extracted features are then fused inside a Transformer that produces a trajectory forecast over a horizon of 6 seconds.

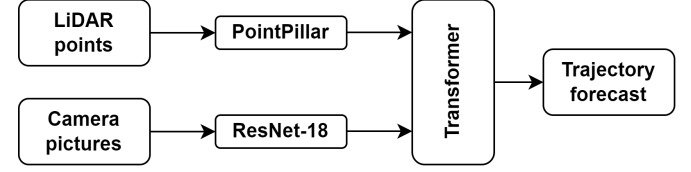


Fig. 1: General structure of the HTTP model.

Figure 1 showcases a block structure of the model and its different components are detailed hereafter.

A. PointPillar

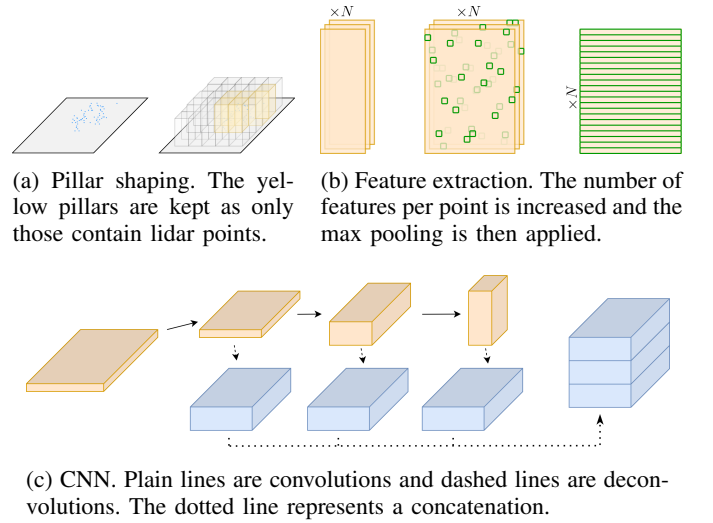


Fig. 2: PointPillar model overview.

The feature extraction of the PointPillar model is split into three successive stages, all shown in fig. 2.

The LiDAR points are first stacked in pillars based on their coordinates. A pillar contains at most 32 points and empty pillars are removed to reduce the memory footprint.

The second stage extracts features from the voxels by adding to each voxel its relative position to the pillar center and the pillar centroid. The number of features is then increased by passing through a two layers network, consisting of a linear operation followed by a batchnorm and a ReLU. Only the maximal feature over the voxels is then kept, leading to 64 features per pillar.

The last stage is a CNN, performing convolutions and deconvolutions on the pillars to extract data dependencies at different resolutions. Note that the pillars from the last stage are first placed on a grid matching the original position of the pillars.

The dimensions of the map on which pillars are placed were determined in order to contain all the voxels in the dataset.

B. Resnet-18

Rather than projecting the raw camera data directly into the target feature dimensions at the encoder, the input passes through a feature extraction network introduced in “Deep Residual Learning for Image Recognition”, known as **Resnet**. Historically, in 2015, image treatment was lead by the assumption that increasing network depth improves in a steady way the feature extraction capacity of the network. However, this idea hits severe optimization bottlenecks. First, the vanishing/exploding gradient during the backpropagation and second, a degradation of the training error once reaching a floor value. The latter degradation reveals a fundamental limitation of such deep network stacking linear transformation and non-linear activation function (e.g. ReLu) which is expressed, in the litterature, by the difficultly of performing an identity mapping for deep layers, or saying otherwise the ability to perform an identity transformation. To give an example, if a 30 layers Convolutional Neural Network (CNN) training leads to a lower error than a 50 layers counterpart, it implies that the 20 additional layers are, in the first place, unable to converge towards the same model accuracy, but furthermore, are distorting the output features by the inability of the NN layers to operate an identity transformation. To overcome these limiting performance constraints, **Resnet** introduces a direct path between the embedded input vector and the output of the feedforward performing a elementwise summation - fig. 3. This path, also known as shortcut connection, makes the build of an identity mapping (no weigth paramater through the direct connection) trivial, while optimizing the features of the **input residual** - $\mathcal{F}(x)$.

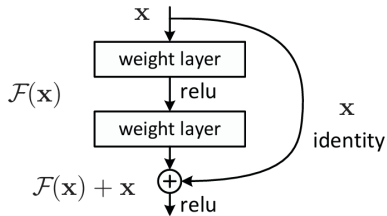


Fig. 3: Resnet block: residual learning.

The HTTP model incorporates a Resnet-18 architecture for image processing. As indicated by the naming, the network contains a total of **18** layers with learnable parameters. The image is first resized, normalized and centered around zero prior to being propagated through the Resnet backbone. The output features are mapped via a linear layer and dimensioned to match the target size required for the embedding vectors at the encoder. Figure 4 showcases the network details.

It is worth noting that while the Resnet-18 has been built from scratch in a dedicated class for learning purpose, the model utilizes the standardized Resnet library enabling the integration of pre-trained weights reducing the training convergence time.

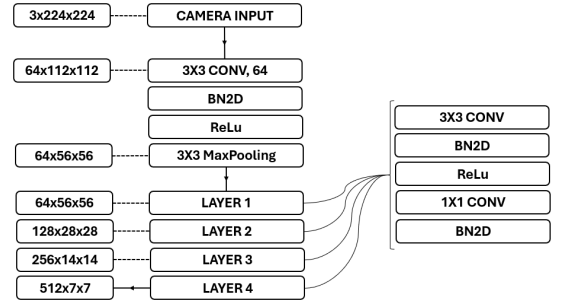


Fig. 4: Resnet18 network architecture.

C. Long Short Term Memory

The last significant data still remaining at our disposal is the past trajectory history of the vehicle whose future path our model attempts to predict. These coordinate sequences carries critical information about the kinetic and momentum state of the vehicle. Nevertheless, these measurements are time-dependent and a standard feedforward linear NN is poorly suited to extract features spanning across distinct timesteps. To efficiently extract these dynamics, our model introduces a Recurrent Neuronal Network block (**RNN**), and more specifically a Long Short Term Memory (**LSTM**) block - [5]. Unlike to a standard RNN relying on a single memory state (known as the hidden state, h_i), the LSTM decouples memory representation into a long term memory cell (C_i) and a short term memory state (h_i).

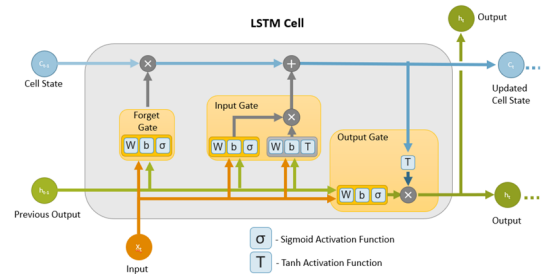


Fig. 5: LSTM network architecture.

The long term memory cell is linked with the short term memory state and the current input through **three gates**, namely the forget gate, the input gate and the candidate gate. These gates regulate how the input and the short term memory modulate the long term memory cell, C_t . It is achieved either by attenuating features by element-wise multiplication with the forgetting output, f_t or by adding new features through the combination of the candidate gate’s output, $\tilde{c}_t$ and input gate’s output, m_t .

$$\begin{aligned}
f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) & (1) \\
\tilde{c}_t &= \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) & (2) \\
m_t &= \sigma(W_m \cdot [h_{t-1}, x_t] + b_m) & (3) \\
C_t &= C_{t-1} \odot f_t + \tilde{c}_t \odot m_t & (4)
\end{aligned}$$

W_h, W_x, W_x, b_f, b_c and b_m denote respectively the weight matrices and the bias vectors associated to the gates, h_{t-1} the previous hidden state and x_t the current input.

The primary advantage of such architecture lies in its stability against the vanishing/exploding gradient descent which is caused, for a standard RNN, by the recursive update of the hidden state at each time step making this long-term temporal dependency sensitive to such gradient problematic.

For the standard RNN, The loss gradient of the hidden state is computed by repeated weight matrix multiplications, W_h , leading to a vanishing or exploding gradient depending on the largest matrix eigenvalue value.

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b) \quad (5)$$

$$\frac{\partial \varepsilon}{\partial h_k} = \frac{\partial \varepsilon}{\partial h_t} \cdot \prod_{j=k+1}^t \text{term}(\tanh(\alpha)) \cdot W_h^T \quad (6)$$

W_h and W_x denote the weight matrices associated to the hidden state and the input, b the bias, ε the error loss and α the term inside the function $\tanh$, eq. (5).

In case of LSTM block, the gradient computation is significantly different and depends on the dynamic activation vector (f_j) programmed by the network. This vector varies in function of the current input and the previous hidden state injected in the block.

$$\frac{\partial \varepsilon}{\partial C_k} = \frac{\partial \varepsilon}{\partial C_t} \cdot \prod_{j=k+1}^t \frac{\partial C_t}{\partial C_{t-1}} \quad (7)$$

$$\frac{\partial C_t}{\partial C_{t-1}} \approx f_t \quad (8)$$

Regarding our model implementation, upstream of the LSTM block, the original past positional history input (4 frames $\times$ 2 coordinates) is first processed by a 2-layer Linear network expanding the 2 coordinates features into 64 then 256 features. The hidden state vector and the long term memory Cell vector are also initialized as a zero vector. Then, the LSTM block is instantiated by the *CustomLSTMCell* class building a feedforward Linear network mapping the concatenated input features dimension and the hidden state vector dimension into a four channel hidden state tensor. The stacking of the weight matrices of each gate into an unified block helps saving time computation and optimizing the GPU usage. By sequentially applying of the gate output equations, the LSTM block returns a post-processed hidden state vector output composed of 256 features.

D. Transformer

The final module and core of our multi-modal model is the encoder-decoder Transformer. As briefly introduced in the state of the art section, the main idea of the transformer is directly inspired from the framework proposed in *DeMo: Decoupling Motion Forecasting into Directional Intentions and Dynamic States* - [6]. Specifically, we adopt the core concept of decoupling the decoder queries into two distinct components: *intent queries*, also referred to as motion or mode queries, mapping the different possible trajectories and the *dynamic queries* inferring the instantaneous momentum state of the target vehicle, which are outputted by our LSTM block.

At the encoder stage, the lidar points and camera pixels are projected into the unified encoder features space dimensions after being pre-processed respectively through the *Pointpillar* and the *Resnet-18* backbones introduced in the previous subsections. The attention head module being permutation invariant (the head is not a sequential process and treat all the tokens in parallel), the network is unaware of the spatial distribution of the tokens and must therefore be explicitly injected into the input embedding vectors. To do so, our model encodes the structural positions of the processed LiDAR and camera outputs using a 2D Sinusoidal Positional Encoding, which are summed with the projected data to form the final attention head inputs.

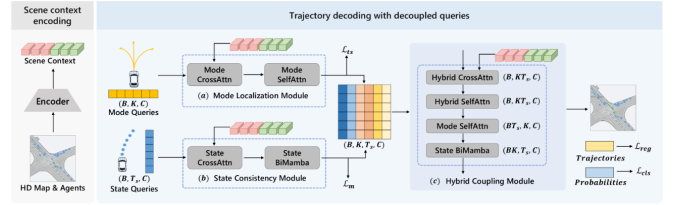


Fig. 6: DeMo model - Transformer architecture.

To meet project deadlines and maintain an appropriate architectural scope for this project, our group opted to reduce the complexity of the Transformer decoder proposed in the DeMo paper - which utilizes two parallel distinct decoders with state and motion queries as input queries finalized with a third fused multi-head attention decoder - Figure 6. While we preserved the core query decoupling strategy, the transformer decoder has been simplified by adapting the standard network structure described in “End-to-End Object Detection with Transformers” - [2] - Figure 7.

The resulting decoder is constructed as follows: the target queries are formed by summing the state query, outputted by the LSTM network which represents the vehicle state content, and the intent queries (M modes), initialized as random distinct vector with modulated variance (σ), matching the targeted features dimension. These intent queries are learnable parameters inferring the spatial intent of the vehicle’s future heading. The combined input vector lies in the space $\mathbb{R}^{M \times 256}$. The output of the decoder is, in fine, mapped to a standard multi-modal trajectory prediction head, used by the DeMo model, projecting decoder embeddings into the future

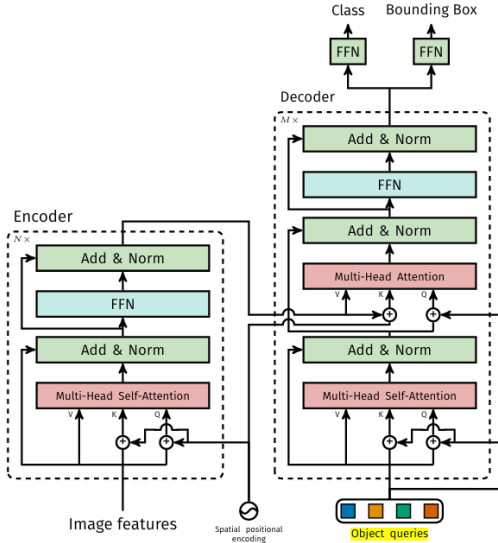


Fig. 7: DETR model - Transformer architecture.

trajectory space, $\mathbb{R}^{M \times 12 \times 2}$ (predicting X and Y displacements over 12 future steps), and the motion mode probability space, $\mathbb{R}^{M \times 1}$.

It is worth noting that the DETR framework investigated the most suitable spatial positioning encoding at the encoder and the decoder stages - Figure 8. Our model is following the proposed strategy with a 2D Sinusoidal encoding at the encoder input and the intent queries summation at each attention layer.

spatial pos. enc. encoder	spatial pos. enc. decoder	output pos. enc. decoder	AP	Δ	AP ₅₀	Δ
none	none	learned at input	32.8	-7.8	55.2	-6.5
sine at input	sine at input	learned at input	39.2	-1.4	60.0	-1.6
learned at attn.	learned at attn.	learned at attn.	39.6	-1.0	60.7	-0.9
none	sine at attn.	learned at attn.	39.3	-1.3	60.3	-1.4
sine at attn.	sine at attn.	learned at attn.	40.6	-	61.6	-

Fig. 8: DETR model - Spatial encoding performance.

E. Inference and Accuracy Metrics

TALK ABOUT Figure 9 AND FINISH CHAPTER

IV. RESULTS

To determine the number of epochs for training, two of the metrics were used: **MinADE** and **MinFDE**. Their respective evolution with the number of epochs is plotted on fig. 10 and the performances start to stagnate from epoch 8, with close to no improvement with an additional training. The model used is therefore the one using 15 epochs, according to the increase in training loss seen in fig. 9.

Figure 11 shows how HTTP predicts the trajectory in different scenarios. With moving cars, the model is quite accurate at least for the first few seconds, as seen in figs. 11a and 11d.

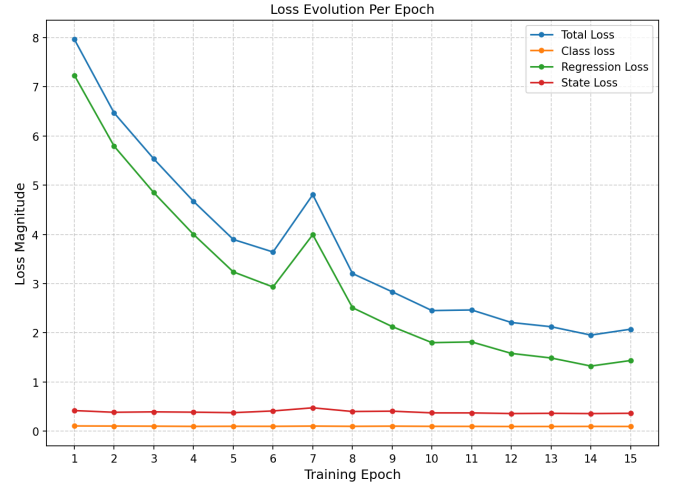


Fig. 9: Evolution of the loss during training.

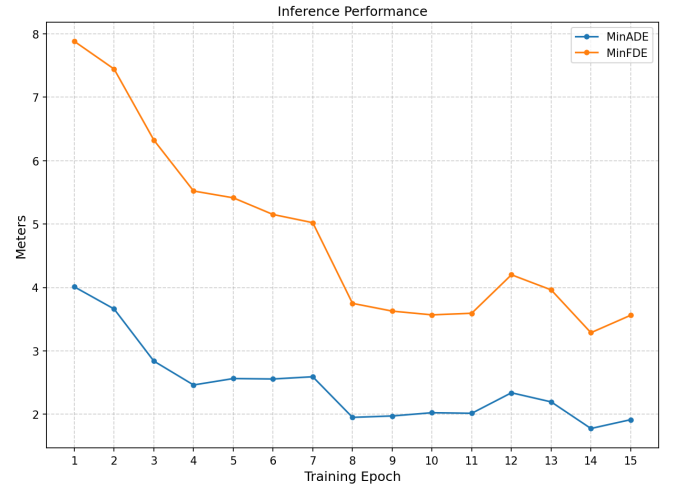


Fig. 10: Evolution of the performances during training.

Stopped cars are much more difficult to handle, as the momentum of the cars is taken into account, making it really hard to understand that a car might start moving. Figure 11b is therefore unable to accurately predict the trajectory of the car that was just stopped. A way to improve this would be to use more situations with traffic lights in the training dataset, or to implement a specific loss function for areas where car have to stop.

Parked cars are accurately predicted, as shown on fig. 11c. The final metrics on the validation dataset with our model are:

- minADE: 1.725 m
- minFDE: 3.337 m
- Miss rate: 18.0%

V. CONCLUSIONS

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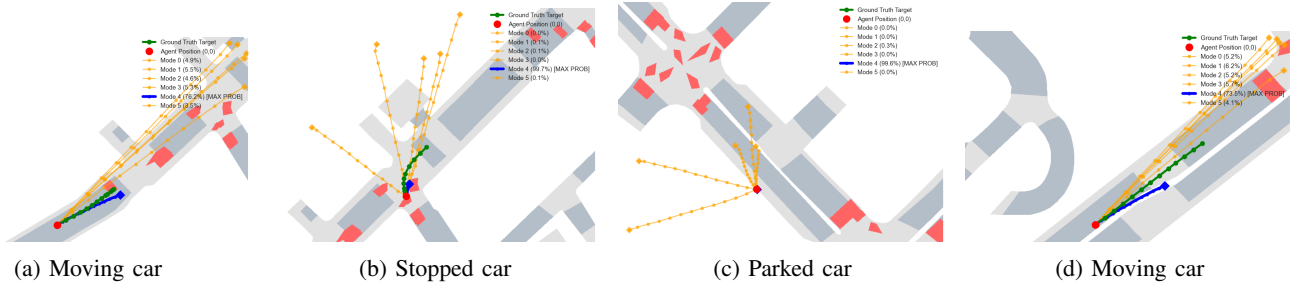


Fig. 11: Overlay of the true trajectory (green) of the cars and the predicted trajectory (blue). The yellow lines correspond to trajectories not selected by the model. Red areas are places where cars should stop.

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